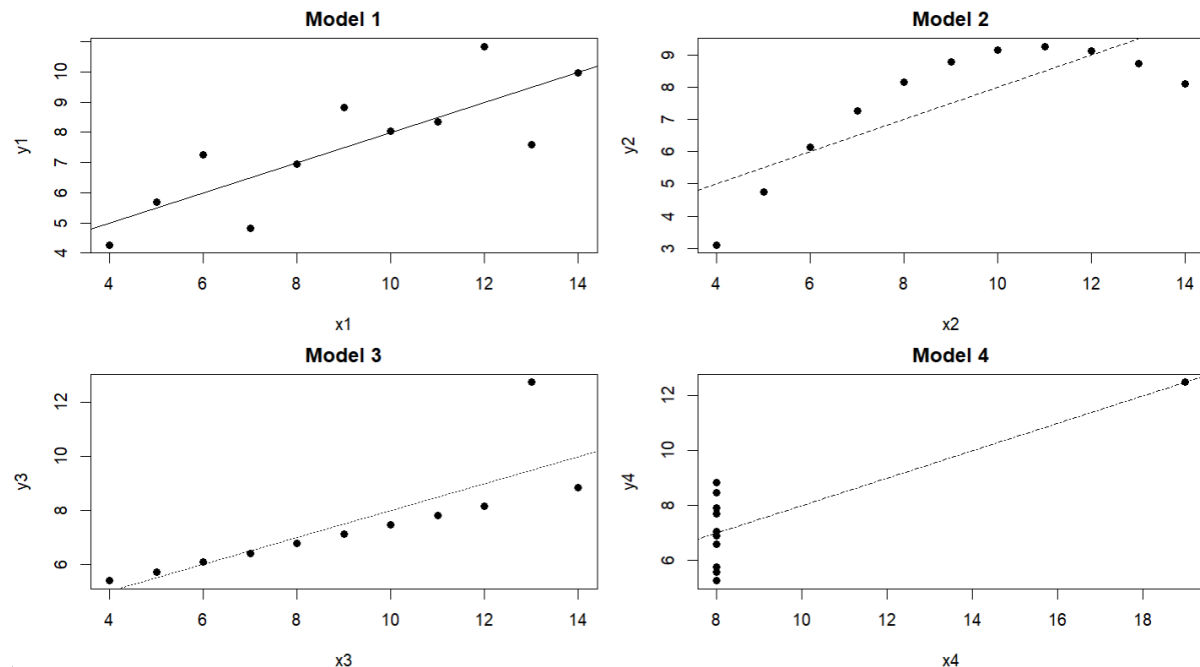


Assignment 03: Part 2. Anscombe's quartet

The Anscombe dataset contains four datasets with paired variables x_1 – x_4 and y_1 – y_4 . I loaded the dataset, examined its contents and summary statistics, and fitted a separate linear regression model for each of the four datasets. The regression results are then used to compare the four models



Regression Model Comparison

Model	Intercept	Slope	R^2	Residual Standard Error
Model 1	3.0001	0.5001	0.6665	1.236
Model 2	3.0009	0.5000	0.6662	1.237
Model 3	3.0025	0.4997	0.6663	1.236
Model 4	3.0017	0.4999	0.6667	1.236

The four regression models produce remarkably similar results. The intercept for all four models is approximately **3**, while the slope is approximately **0.5**. The R^2 values are also almost identical, at approximately **0.67**, indicating a similar amount of variation explained by the linear models. The residual standard error is approximately **1.24** for all four models.

Based only on these regression statistics, the four datasets appear to have almost the same linear relationship. However, the similarity of these numerical results does not mean that the four datasets have the same underlying pattern. This is the main reason why visualization is important when analyzing data.

1. Model 1 — Consistent Linear Pattern

Model 1 shows a relatively clear **linear relationship**. The points are distributed around a straight-line pattern, and the regression line passes through the general centre of the observations. No single observation appears to strongly determine the relationship.

The numerical regression results, including the slope of approximately 0.5 and R^2 of approximately 0.67, provide a reasonably useful summary of the pattern visible in the plot. However, the scatterplot is still important because it allows us to visually examine whether the linear model is appropriate.

2. Model 2 — Curved Pattern

Model 2 shows a **curved relationship** rather than a straight-line pattern. Although the regression model produces almost exactly the same slope, intercept, R^2 , and residual standard error as Model 1, the straight regression line does not adequately describe the actual shape of the data.

The numerical summary cannot reveal this nonlinear structure. An R^2 of approximately 0.67 does not indicate whether the observations follow a straight line, a curve, or another pattern. The scatterplot therefore reveals an important feature that is hidden by the regression statistics. A nonlinear model or an appropriate transformation may be more suitable for this type of relationship.

3. Model 3 — Linear Pattern with an Outlier

Model 3 shows a fairly clear **linear pattern for most of the observations**, but one observation is noticeably separated from the main group. This observation is an **outlier**.

The regression statistics are very similar to those of the other datasets and do not clearly reveal the presence of this unusual observation. The scatterplot makes the outlier immediately visible.

This demonstrates why graphical examination is important in regression analysis. The observation should not automatically be removed; instead, it should be investigated to determine whether it is a valid observation or a possible data error and to understand its effect on the regression results.

4. Model 4 — Relationship Influenced by One Observation

Model 4 shows that most observations are concentrated around a relatively narrow range of x-values, while one observation is located far away from the others in the x-direction. This observation has **high leverage** because of its unusual x-value.

The observation strongly influences the fitted regression line and helps create the apparent linear relationship. However, the regression statistics are almost identical to those of the other three datasets and do not reveal how strongly the relationship is influenced by this single observation.

The scatterplot makes this feature immediately visible. It demonstrates that numerical regression results alone may hide influential observations and that graphical examination is necessary to understand the actual structure of the data.